

2022-Ph-H1-Q7

Section: Our Dynamic Universe

Topic: Forces, Energy and Power

A lift is accelerating upwards. What can be said about the tension in the cable?

Answer: D

(Tension is greater than the weight of the lift.)

Guidance for Students:

Upward acceleration requires a **net upward force**. So the tension must exceed the downward force of gravity.

Revision Tips:

- Upwards acceleration $\rightarrow$ tension $>$ weight.
- Net force $= T - W$
- Use $F = ma$ if needed.